

DATABRICKS · HEALTHCARE / ICU OPERATIONS

ICU Step-Down Readiness

Turning scarce ICU capacity into a decision advantage — an AI-assisted way to see, every shift, which patients are most ready to safely leave intensive care.

Lakeflow

Unity Catalog

Lakebase

Mosaic AI + Foundation Models

Genie

Databricks Apps

THE PROBLEM

A blocked ICU bed is a clinical and financial bottleneck

70–90%

Typical ICU occupancy
— one blocked bed
delays a critical
admission

Manual

Step-down decisions
are subjective and vary
by provider and shift

**"To be
safe"**

Clinicians hold patients
longer than needed,
lengthening stays

The question every intensivist asks on rounds — *“who can safely move to step-down today?”*
— has no consistent, data-driven answer.

THE OUTCOME WE SELL

Free ICU capacity without compromising safety

**10–
15%**

Target
reduction
in ICU
length of
stay

**\$2–
5k**

Saved per
accelerated
discharge-
day

+beds

More ICU
capacity for
emergent
admissions

Consistency

Evidence-informed
decisions across shifts

For a 30-bed ICU, a 0.8-day average LOS reduction is on the order of **hundreds of additional patient-days of capacity per year**.

WHAT WE BUILT

One governed journey — raw data to a clinician's screen



Integrated, not siloed: the same governed gold tables feed the operational app, the model, and natural-language querying — one lineage, one source of truth.

THE PRODUCT

A ranked ICU census the care team acts on

- **Census dashboard** — every current patient ranked by a relative readiness index, color-banded Ready / Borderline / Not-ready.
- **Patient detail** — the factors driving each recommendation, vital-sign trends, and a plain-language **AI clinical summary**.
- **Analytics** — capacity and outcome trends, with honest model performance.

Live: [icu-step-down-...
databricksapps.com](#)

Example — ranked census

- **92** · Ready · off pressors 36h, GCS 15
- **78** · Ready · extubated, stable VS
- **54** · Borderline · lactate ↑
- **18** · Not ready · on vent + pressors

FOR THE CMIO & ICU MEDICAL DIRECTOR

Built to be trusted, governed, and operated

Governed by design

Unity Catalog across the medallion — PII classification tags on identifiers, least-privilege grants, and full bronze→gold lineage.

Operationally served

Gold features synced to Lakebase (Postgres) for low-latency app reads; model served on Mosaic AI; narrative via the Foundation Model API.

Reproducible

Everything deploys via Databricks Asset Bundles; 117 backend + 32 frontend + 17 pipeline-integration tests.

Safe by framing

De-identified public data; a relative decision-support index, not an automated discharge gate.

MODEL HONESTY

Decision support, not a discharge switch

AUC 0.66

Honest, modest discrimination on the retrospective feature set

Index, not %

Relative readiness rank within the current census — never a "probability of safe discharge"

Explainable

SHAP factors in clinical language; known risks always shown as risks

The roadmap to stronger discrimination: richer real-time features, calibration, and prospective validation with the clinical team.

THE ASK

From prototype to a unit pilot

- **VP of Clinical Operations / CMO** — measured on ICU length of stay, bed occupancy, and emergent-admission wait times: sponsor a 60–90 day pilot in one ICU, scored on LOS reduction with the safe-transfer rate held or improved.
- **ICU Medical Director** — measured on safe-transfer rate and 48–72h ICU bounce-back: co-define the readiness features and validate the index against your step-down criteria.
- **CMIO / VP Data & Analytics** — measured on time-to-value and governed reuse: point the same pipeline at your EHR feed — the architecture is unchanged.

Every ask maps to a number already on that leader's own scorecard. The capability is live today on Databricks; the next step is your data and your clinicians.